

## S4 Supplementary Clinical Summary

*A cautious clinician-facing interpretation of the article*

Short title: S4 Clinical Summary - NeuralManifoldDynamics

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### Scope and intent

#### **Important: observational study - no intervention tested**

The study tested no clinical intervention. Participants performed laboratory tasks or contributed previously collected recordings; no physiology or brain state was manipulated directly. Findings are observational, cross-sectional, and group-level. No individual-level predictions or causal conclusions are supported.

This supplement translates the measurement contract for clinically oriented readers without amplifying it into a diagnostic claim. The main contribution is software and measurement infrastructure: it makes modality-specific feature surfaces, low-dimensional proxy trajectories, quality flags, and provenance available for later research. Reference and validation cohorts are not interchangeable, and some are healthy laboratory samples rather than patient cohorts.

## What is the framework, and why does it matter here?

The framework converts a window of neural or physiological data into a reproducible vector of features, then applies a fixed versioned mapping to a three-dimensional proxy chart and an optional nine-dimensional refinement. This can help clinical researchers compare processing choices, identify unsupported windows, and preserve the exact provenance of a derived result.

The words **mobility**, **deviation**, and **entropy** describe operational proxy families in this release. They do not prove that a patient occupies a unique biological state or that the chart is a validated clinical biomarker.

## What was studied

The repository contains reference EEG and fMRI configurations, plus validation examples using EEG with ECG/PPG/pupil recordings, simultaneous MEG/EEG, sleep data, verbal working memory, Parkinson-related recordings, DANDI/NWB neurophysiology, and Figshare BDF infant EEG. These streams primarily test whether data can be indexed, transformed, quality-controlled, and serialized under one auditable contract.

The ds003838 export completed for 130 subjects; the corrected stage-contrast statistics used 62 subjects with complete stage coverage. The ds003645 MEG result is a five-subject pilot with readiness 0.7879. DANDI LFP and infant EEG results are transport/QC demonstrations rather than clinical endpoint studies.

## What the coordinates mean in plain language

- **m** is a release-fixed proxy family emphasizing slow-wave, alpha, and macrostate-related feature changes.

- `d` is a proxy family emphasizing dispersion, high-frequency or binding features, and local dynamical variation.
- `e` is a proxy family emphasizing nonlinear complexity and an additive arousal/emodied fallback surface.
- `mnp3_3d_dot` describes change along the proxy chart; it is not a clinical rate of disease progression.
- A Jacobian describes local deformation of the proxy trajectory. It is useful for auditing local geometry but is not a direct physiological mechanism.
- `anchor_state` contains time-aligned body-state proxies such as HRV-derived indices. It is parallel to the neural chart, not a fourth neural coordinate.

## The main findings, in order of clinical salience

### 1. The contract improves auditability

The strongest result is infrastructural. The pipeline records coverage, invalidity, feature transforms, source rows, anchors, and optional Jacobians in self-describing outputs. This can make future clinical analyses easier to reproduce and easier to reject when support is inadequate.

### 2. The ds003838 physiology surface changes across task stages

In the corrected ds003838 validation package, `vagal_index`, sympathetic and HRV measures, trajectory path length, and related coupling measures varied across rest, listening, and memory-load stages. The listen–mem13 `vagal_index` contrast had Cohen’s  $d = 1.995$  and Wilcoxon  $p = 1.2 \times 10^{-10}$  on the complete-case sample  $N = 62$ . This is an internal task-contrast validation of the export surface, not evidence of a clinical autonomic treatment effect.

### **3. MEG is usable as a pilot mapping, not as a clinical modality claim**

The ds003645 pilot showed task-sensitive face/scrambled separation across several window sizes. Its readiness score was 0.7879, below the 0.80 expansion threshold, and subjects showed a bimodal d-family pattern. A later holdout did not reproduce matched EEG–MEG residual topography. The appropriate clinical reading is that the engineering path is promising enough for further study, not that MEG and EEG have been shown to share one validated manifold.

### **4. Neuropixels/LFP ingest is a transport demonstration on animal data, not a human clinical finding**

The DANDI 000006 (spike-sorted Units) and 000458 (LFP) examples are rodent recordings used to show that spike-rate and continuous-electrode data can be ingested, quality-controlled, and mapped into the same measurement contract as the human EEG/MEG/fMRI pipelines. No readiness score, no cross-species comparison, and no clinical or circuit-level claim accompanies these examples. They should be read as an ingest-format and QC extension only.

### **What was NOT found - equally important**

The repository does not establish diagnostic accuracy, individual prediction, causal treatment response, or superiority over HMM, microstate, CAP, LEiDA, CEBRA, or other latent-state methods. The ds003838 effect has no independent cohort replication in this evidence pack. Working-memory HRV claims in ds006848 were withheld because 60 s windows were contaminated by retrieval events. Pupillometry is implemented but not validated at dataset scale. DANDI LFP contact/reference comparisons are descriptive one-session checks, and infant BDF Status labels do not support rest/reaching condition contrasts.

## **How to read the technical terminology clinically**

“Proxy” means a fixed computational summary, not a validated biomarker. “Anchor” means a time-aligned physiological surface, not proof that the body signal causes a neural coordinate. “Geometry validity” means that numerical support and conditioning passed the stated checks, not that the underlying biology is correct. “Internal validation” means the repository contains a reproducible run or audit; it does not mean independent replication.

## **What clinicians should not conclude**

Readers should not infer diagnosis, prognosis, consciousness level, treatment response, patient-level risk, or causal physiology from these outputs. They should not use `m`, `d`, `e`, `vagal_index`, or a readiness score as a clinical decision threshold. They should not interpret a high-quality HDF5 file as evidence that the chosen proxy chart is biologically unique.

## **What would increase clinical confidence**

Confidence would require preregistered replication in independent patient samples, longitudinal within-person designs, validated clinical outcomes, explicit out-of-sample evaluation, modality-specific artifact cleaning, calibration of individual-level uncertainty, and intervention studies. Chart stability under feature substitution and projection perturbation should also be tested before clinical interpretation.

## **Summary for clinical readers**

NeuralManifoldDynamics is best treated as an auditable research measurement contract. It can organize multimodal data into versioned proxy trajectories, physiological sidecars, geometry

diagnostics, and provenance-rich artifacts. Some task-linked internal validation results are encouraging, especially the corrected ds003838 stage package, while MEG, LFP, infant EEG, and phase-anchor lines remain exploratory or transport-focused. Nothing in the current evidence supports diagnostic use, individual prediction, or causal clinical claims.

## **Bibliography**

## **Competing Interests**

The authors have declared that no competing interests exist.